Setting a Paragraph Whole

Optimum-fit line breaking in a troff for InferNode

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Abstract A paragraph broken one line at a time is broken greedily: each line takes as many words as fit, and a loose line is discovered only when it can no longer be helped. Knuth and Plass showed that choosing every break of a paragraph at once, as the shortest path through a graph of feasible breaks, gives evenly spaced text with fewer hyphens. We describe how the method fits troff's model of filling and adjusting, and compare its spacing with LuaLaTeX's on the same prose.

Keywords typesetting, line breaking, troff, dynamic programming

1 Introduction

Troff fills a line until the next word will not fit, then adjusts it to the measure by widening its spaces [1]. The rule is simple and fast, and it is local: the line just set is never reconsidered. When a long word follows a short line, the line must be stretched however far it takes, and the reader sees a river of white running down the page.

TeX instead treats a paragraph as a whole [2]. Every place a line may end is a node; every way of setting the words between two such places as a line is an edge, weighted by how badly the line would look. The best paragraph is the cheapest path from the first node to the last, found by dynamic programming in time linear in the length of the paragraph when, as in practice, lines are few words long.[1]

2 The cost of a line

A line's spaces can stretch or shrink. Let r be the ratio by which they must, as a fraction of what they allow. Its badness grows as the cube of that ratio, and a line's demerits combine it with the penalty p of the break it ends at:

$$d = \left(l + 100|r|^3\right)^2 + p^2, \tag{1}$$

where l is a small constant that makes fewer lines better than more. A hyphen's break carries a penalty, and two hyphenated lines in a row an extra one, so hyphens are few and seldom stacked.

Figure 1 shows the idea for a paragraph of three lines. Breaking greedily would take the first arc that fits and follow it; the whole-paragraph method keeps, for each node, only the cheapest way of reaching it, and so considers every path without enumerating them.

1 A line can end only where it would not be stretched or shrunk beyond what its spaces allow, so each node has only a handful of predecessors.

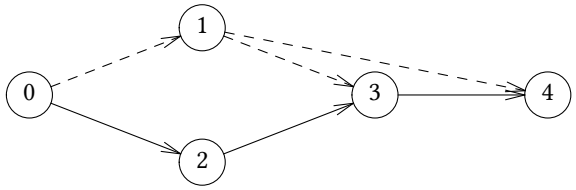


Figure 1. Feasible breaks of a short paragraph. Each arc is a line; the solid path is the one of least total demerits.

3 Fitting it to troff

Troff's filling is driven by its input: words arrive one at a time, and requests such as a break or a change of font may come between them. Our troff collects a paragraph's words, with their fonts, sizes and the escapes between them, until something ends the paragraph, and only then chooses its breaks.

Two of TeX's refinements carry over directly. A space may shrink by a third, so that a line need not be broken short and stretched; and a hyphen costs as a loose line does, a second in a row ten times as much, a fourth not allowed at all.

4 Results

We set 180 pages of prose, the documentation of this system, with our troff and with LuaLaTeX and its microtype package, on the same page and in the same face, and measured every justified line of each. Table 1 gives the results.

	troff	LuaLaTeX
mean space (em)	0.27	0.28
95th percentile (em)	0.38	0.35
lines looser than 0.5 em	21	25
lines hyphenated	10%	8%
two hyphens in a row	32	28

Table 1. Interword spacing and hyphenation on 180 pages of prose.

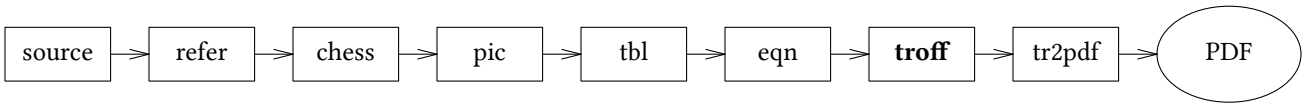


Figure 2. The pipeline book(1) runs. Each preprocessor runs only if the document uses it; troff runs as often as cross-references need.

The two are close. Troff's spaces are as even on average, with a slightly longer tail; it hyphenates a little more often.

5 Pictures and the rest

Figures in this article are drawn with pic [3], equations set by eqn [4] and tables by tbl. Each is a preprocessor that turns its own small language into troff's requests, so that the formatter itself need know nothing of graphs, formulas or tables; the pipeline that runs them is shown in figure 2. The same arrangement serves an article and a book: the class decides the page, and the tools are shared.

6 Conclusion

Whole-paragraph line breaking costs troff little: a page of prose is set in about twenty milliseconds, and its spacing is as even as LaTeX's. The method belongs as much to troff's tradition of small, composable programs as to TeX's.

Acknowledgements

The measurements were made with the tools described here, on the documents distributed with them.

References

[1] J. F. Ossanna, "NROFF/TROFF User's Manual," Bell Laboratories, Murray Hill, N.J., Computing Science Technical Report 54, 1976.

[2] D. E. Knuth and M. F. Plass, "Breaking Paragraphs into Lines," *Software: Practice and Experience*, vol. 11, no. 11, pp. 1119–1184, 1981.

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[4] B. W. Kernighan and L. L. Cherry, "A System for Typesetting Mathematics," *Communications of the ACM*, vol. 18, no. 3, pp. 151–157, 1975.